Reducing Input Size

Case Study: Clustering (Part 2)

OUTLINE

1. k-center clustering
   - Sequential Farthest-First Traversal.
   - Coreset technique.
   - MapReduce Farthest-First Traversal.
2. Coreset-based max pairwise distance approximation.

k-center clustering

Farthest-First Traversal: algorithm

- Popular 2-approximation sequential algorithm developed by T.F. Gonzalez [Theoretical Computer Science, 38:293-306, 1985]
- Simple and, somewhat fast, implementation when the input fits in main memory.
- Powerful primitive for extracting samples for further analyses

Input Set P of N points from a metric space (M, d), integer $k > 1$

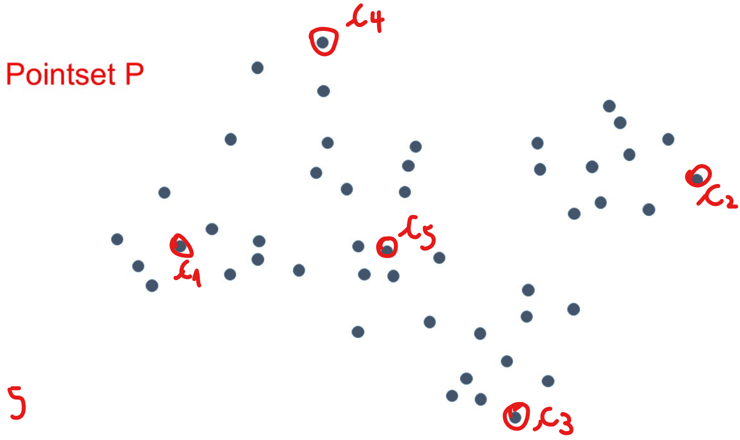
Output k-clustering $\mathcal{C} = (C_1, C_2, \ldots, C_k; c_1, c_2, \ldots, c_k)$ of P which is a good approximation to the k-center problem for P.

Farthest-First Traversal: algorithm

$S \leftarrow \{c_1\}$ `//` $c_1 \in P$ *arbitrary point*
for $i \leftarrow 2$ **to** k **do**
  `Find the point` $c_i \in P - S$ `that maximizes` $d(c_i, S)$
  $S \leftarrow S \cup \{c_i\}$
return `Assign`(P, S)

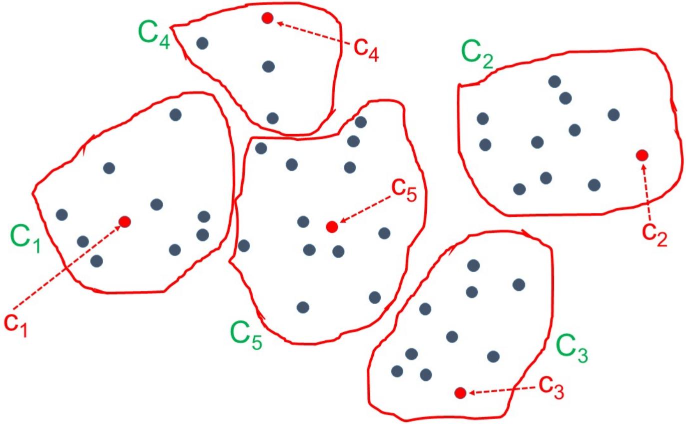
Observation: The invocation of Assign(P, S) at the end can be avoided: *in every iteration of the for-loop maintain, for each point, its closest current center*.

Farthest-First Traversal: example ($k = 5$)



$K = 5$

Farthest-First Traversal: example ($k = 5$)



Exercise

Show that the Farthest-First Traversal algorithm can be implemented to run in $O(N \cdot k)$ time.

Hint: make sure that in each iteration i of the for-loop each point $p \in P - S$ knows its closest center among $c_1, c_2, \ldots, c_{i-1}$ and the distance from such a center.

Farthest-First Traversal: analysis

Theorem

Let $\mathcal{C}_{\text{alg}}$ be the k-clustering of P returned by the Farthest-First Traversal algorithm. Then:

$$\Phi_{\text{kcenter}}(\mathcal{C}_{\text{alg}}) \leq 2 \cdot \Phi^{\text{opt}}_{\text{kcenter}}(P, k).$$

That is, Farthest-First Traversal is a 2-approximation algorithm.

Proof of Theorem

$\mathcal{C} = (C_1, C_2, \ldots, C_k ; \underbrace{c_1, c_2, \ldots, c_k}_{S})$

$S = \{c_1, c_2, \ldots, c_k\}$ indices based on order of discovery

Let q = farthestmost point of $P - S$ from S

$$d(q, S) = \max_{x \in P - S} d(x, S)$$

Proof of Theorem

Observe that

* $d(q, S) = \Phi_{kcenter}(\mathcal{C})$ by definition

* q would be the next center selected by Farthest-First Traversal if a k'-clustering of P, with $k' > k$, were sought (in this case $q = c_{k+1}$)

Proof of Theorem

The selection procedure ensures that

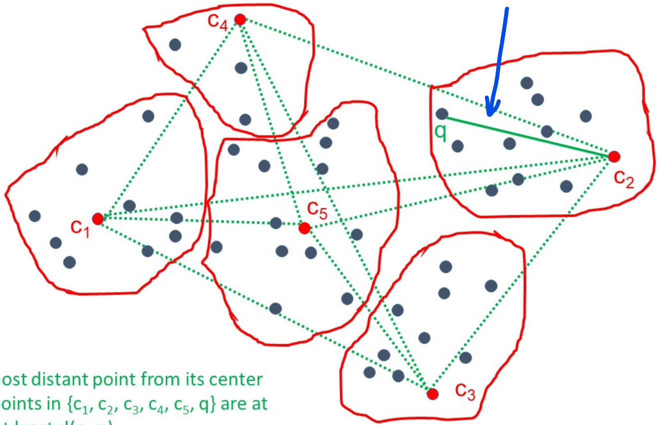
$$d(c_2, c_1) \geq$$
$$d(c_3, \{c_2, c_1\}) \geq$$
$$d(c_4, \{c_3, c_2, c_1\}) \geq$$
$$\vdots$$
$$d(c_i, \{c_{i-1}, \ldots, c_1\}) \geq$$
$$\vdots$$
$$d(c_k, \{c_{k-1}, \ , c_1\}) \geq$$
$$d(q, \{c_k, \ldots, c_1\}) = d(q, S) = \Phi_{kcenter}(\mathcal{P})$$

Proof of Theorem

Now, the set $\{c_1, c_2, \ldots, c_k, q\}$ contains $k+1$ distinct points. Moreover, the sequence of inequalities in the previous slide implies immediately that any 2 points from $\{c_1, c_2, \ldots, c_k, q\}$ are at distance $\geq \Phi_{kcenter}(\mathcal{C})$ from one another (A)

Proof of Theorem

$d(q,S) = \Phi_{kcenter}(C)$



- q is the most distant point from its center
- any two points in $\{c_1, c_2, c_3, c_4, c_5, q\}$ are at distance at least $d(c_2,q)$

Proof of Theorem

Consider an optimal k-clustering for P w.r.t. $\Phi_{kcenter}$

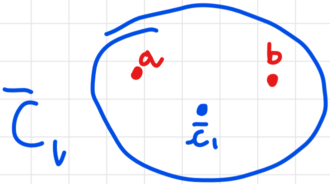
$$\bar{\mathcal{C}} = (\bar{C}_1, \bar{C}_2, \ldots, \bar{C}_k; \underbrace{\bar{c}_1, \bar{c}_2, \ldots, \bar{c}_k}_{\bar{S}})$$

* Since $\bar{\mathcal{C}}$ is optimal w.r.t $\Phi_{kcenter}$, we have

$$\max_{x \in P} d(x, \bar{S}) = \Phi_{kcenter}(\bar{\mathcal{C}}) = \Phi^{opt}_{kcenter}(P, k)$$

* Since $\{c_1, c_2, \ldots, c_k, q\}$ contain $k+1$ distinct points of P there must exist 2 of these points, say a and b, which belong to the same optimal cluster, say $\bar{C}_i$

Proof of Theorem



Thus:

$$
\begin{aligned}
\Phi_{kcenter}(\mathcal{C}) &\leq d(a,b) && \text{(by property A)} \\
&\leq d(a,\bar{c}_1) + d(\bar{c}_1, b) && \text{(by triangle inequality)} \\
&\leq 2\,\Phi_{kcenter}(\bar{\mathcal{C}}) \\
&= 2\,\Phi^{OPT}_{kcenter}(P,k) && \text{(by optimality of } \bar{\mathcal{C}}\text{)}
\end{aligned}
$$

$\square$

Proof of Theorem

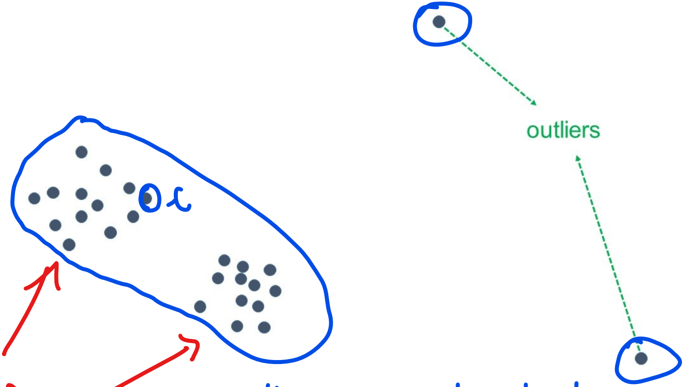
Obs The proof crucially exploits the fact that the centers are "well spaced" from one another, a property that other 2-approximation algorithms for k-center clustering do not have, which can be useful for other applications (e.g. DIVERSITY MAXIMIZATION)

Observations on k-center clustering

- The k-center objective focuses on worst-case distance of points from their closest centers.
- Farthest-First Traversal's approximation guarantees are almost the best one can obtain in practice. It was proved that computing a c-approximate solution to k-center is NP-hard for any fixed $c < 2$.
- The k-center objective is very sensitive to noise. For noisy datasets (e.g., with outliers), the clustering which optimizes the k-center objective may obfuscate some "natural" clustering inherent in the data. See example in the next slide.

Example: noisy pointset

what is the best 3-clustering for this pointset



2 natural clusters

with a 3 center budget we are unable to separate the natural clusters

k-center clustering for big data

Observation. Farthest-First Traversal requires $k-1$ scans of the pointset P with P stored in main memory: impractical for massive P and k not so small.

How can we compute a "good" k-center clustering of a pointset P that is too large for a single machine?

We may resort to the (composable) coreset technique.

Coreset technique

Suppose that we want to solve a problem Π for a massive input P.

Coreset technique

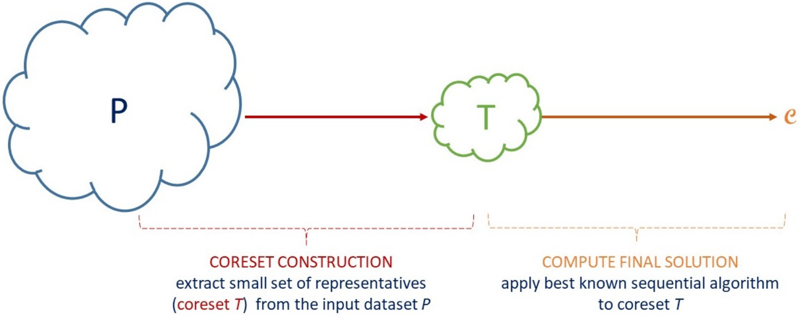
1. Extract a "small" subset T from P (dubbed coreset), making sure that it represents P well. (for solving Π)
2. Run best known (sequential) algorithm for Π, possibly expensive, on the small coreset T, rather than on the entire input P.

The technique is effective if

- T can be extracted efficiently from P.
- The solution computed on T is a good solution for Π w.r.t. the entire input P.

Coreset technique

Obs uniform random sampling is a way to extract a coreset but, in some cases, not the most effective one



Composable Coreset technique

Composable Coreset technique

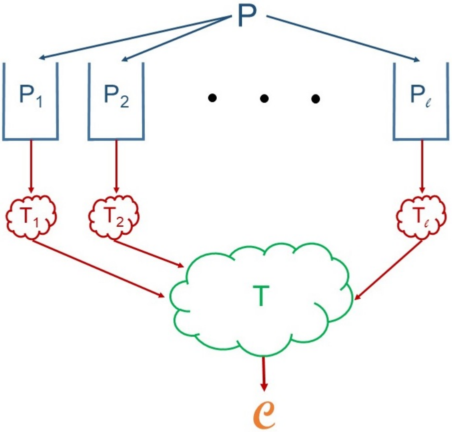
1. Partition P into ℓ subsets $P_1, P_2, \ldots, P_\ell$, and extract a "small" coreset T_i from each P_i, making sure that it represents P_i well. T_i's can be extracted in parallel
2. Run best known (sequential) algorithm for Π, possibly expensive, on $T = \cup_{i=1,\ell} T_i$, rather than on P.

$T \equiv$ composition of coresets

The technique is effective if

- Each T_i can be extracted efficiently from P_i (in parallel for all i's).
- The final coreset T is still small and the solution computed on T is a good solution for Π w.r.t. the entire input P.

Composable coreset technique



- Extract a *small coreset* T_i from each P_i
- Apply best known sequential algorithm to final coreset T = union of the T_i's, to compute final solution $\mathcal{C}$

Application to k-center clustering

Main ideas

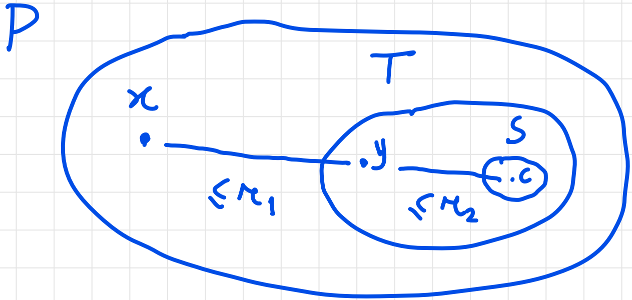
* Input instance (P, k)
* (1) Extract $T \subseteq P$ $\quad (T = \bigcup T_i \quad T_i \subseteq P_i)$

  $P = P_1 \cup P_2 \cup \quad \cup P_l$

  $\forall x \in P \quad d(x, T) \leq r_1$ (small)
* (2) Compute the final set S of k centers from T

  s.t. $\forall y \in T \quad d(y, S) \leq r_2$ (small)
* Use Farthest-First Traversal for both (1) and (2)

Application to k-center clustering



$$d(x, S) \leq d(x, c) \leq r_1 + r_2$$

Our strategy will assume that

$$r_1, r_2 \leq 2\,\Phi^{OPT}_{kcenter}(P, k) \Rightarrow d(x, S) \leq 4\,\Phi^{OPT}_{kcenter}(P, k)$$

$\Rightarrow$ 4-approximation

MapReduce-Farthest-First Traversal

Let P be a set of N points (N large!) from a metric space (M, d), and let $k > 1$ be an integer.

Exercise: fill in details (e.g. key-value representation)

Algorithm MR-Farthest-First Traversal

- **Round 1:**
  - Map Phase: Partition P arbitrarily into ℓ subsets of equal size $P_1, P_2, \ldots, P_\ell$.
  - Reduce Phase: for every $i \in [1, \ell]$ separately, run Farthest-First Traversal on P_i to determine a set $T_i \subseteq P_i$ of k centers.
- **Round 2:**
  - Map Phase: empty.
  - Reduce Phase: gather the coreset $T = \cup_{i=1}^{\ell} T_i$ (of size $\ell \cdot k$) and run, using a single reducer, Farthest-First Traversal on T to determine a set $S = \{c_1, c_2, \ldots, c_k\}$ of k centers.

  saved into a global variable
- **Round 3:** Execute Assign(P, S).

Observations

- In Rounds 1 and 2 the Farthest-First Traversal is used to determine only centers and not complete clusterings.
- Farthest-First Traversal is used both to compute the coresets T_i (hence the final coreset T) and to compute the final set of centers.

Representation

Input: $\{(ID_x, x) \quad x \in P \quad ID_x \in [0, N)\}$

Output: $\{(ID_x, (x, c)) : x \in P \quad c \in S$ closest center to x, $ID_x \in [0, N)\}$

Analysis of MR-Farthest-First Traversal

Assume $k = o(N)$ e.g. $k \leq \sqrt{N}$

3 Rounds

Aggregate space $M_A = O(N)$ no replication of data but only selection of T_i's, S

local space M_L

* Round 1 : $O(N/\ell)$ to process each P_i separately

Analysis of MR-Farthest-First Traversal

* Round 2 : $O(\ell\ k)$
* Round 3 : $O(k)$ to store S

$\Rightarrow M_L = O(\max\{N/\ell,\ \ell \cdot k\})$

Best choice for ℓ: $\ell = \sqrt{N/k}$

$\Rightarrow M_L = O(\sqrt{N \cdot k}) = o(N)$ if $k = o(N)$

Typically, k is very small, so $M_L \ll N$

Analysis of MR-Farthest-First Traversal

Let T be the union of the coresets T_i computed by MR-Farthest-First Traversal on input P. The following lemma establishes the quality of T.

Lemma

For every $x \in P$ we have

$$d(x, T) \leq 2 \cdot \Phi^{\mathrm{opt}}_{\mathrm{kcenter}}(P, k).$$

$\Rightarrow$ T is a good coreset

Obs In this case uniform sampling to select T would not work well

If B is small it may be not represented in the sample

P

$|B| = o(|A|)$

Proof of Lemma

Focus on P_i $\quad (P_i \subseteq P)$

* T_i = centers extracted from P_i using Farthest-First Traversal ($|T_i| = k$)
* q_i = point of P_i with max distance from $T_i \Rightarrow d(q_i, T_i) = \max_{x \in P_i} d(x, T_i)$

$T_i \cup \{q_i\}$ is a set of $k+1$ points from P_i (hence from P)
By reasoning as in the analysis of Farthest-First Traversal we can show that any 2 points of $T_i \cup \{q_i\}$ are at

Proof of Lemma

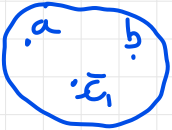
distance $\geq d(q_1, T_1)$ from one-another $\Rightarrow$

there must be a cluster $\bar{C}_i$ from the optimal

k-clustering $\bar{\mathcal{C}} = (\bar{C}_1, \bar{C}_2, \dots, \bar{C}_k; \bar{c}_1, \bar{c}_2, \dots, \bar{c}_k)$ of P

that contains 2 points of $T_1 \cup \{q_1\}$, say a and b

$\bar{C}_1$



$$
\begin{aligned}
d(q_i, T_1) &\leq d(a, b) \\
&\leq d(a, \bar{c}_1) + d(\bar{c}_1, b) \\
&\leq 2\,\Phi^{OPT}_{kcenter}(P, k)
\end{aligned}
$$

Proof of Lemma

Consider an arbitrary $x \in P$

WLOG: assume $x \in P_1$

$$d(x, T) \leq d(x, T_1) \quad (\text{since } T_1 \subseteq T)$$

$$\leq d(q_1, T_1) \quad (\text{by choice of } q_1)$$

$$\leq 2\,\Phi^{OPT}_{kcenter}(P, K)$$

$\square$

$$\forall x \in P \ \exists y \in T \cdot d(x, y) \leq 2\,\Phi^{OPT}_{kcenter}(P, K)$$

Analysis of MR-Farthest-First Traversal

Theorem

Let $\mathcal{C}_{\mathrm{alg}}$ be the k-clustering of P returned by the MR-Farthest-First Traversal algorithm. Then:

$$\Phi_{\mathrm{kcenter}}(\mathcal{C}_{\mathrm{alg}}) \leq 4 \cdot \Phi^{\mathrm{opt}}_{\mathrm{kcenter}}(P, k).$$

That is, MR-Farthest-First Traversal is a 4-approximation algorithm.

Proof of Theorem

$S \equiv$ centers extracted from T using Farthest-First Traversal

$q \equiv$ point of T which is at max distance from S

$$\Rightarrow d(q,S) = \max_{y \in T} d(y,S)$$

By repeating the same argument as before we can show that $d(q,S) \leq 2\Phi^{OPT}_{kcenter}(P,S)$

$$\Rightarrow \forall y \in T \quad d(y,S) \leq d(q,S) \leq 2\Phi^{OPT}_{kcenter}(P,K)$$

Proof of Theorem

$\Rightarrow \forall y \in T \quad \exists c \in S \; : \; d(y,c) \leq 2\Phi^{OPT}_{kcenter}(P,k)$

By combining this with the result of the lemma we conclude that

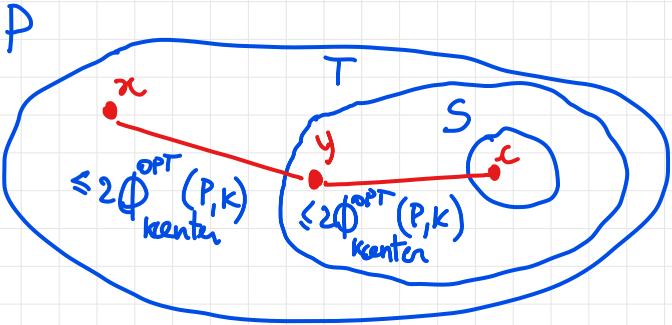
$\forall x \in P \quad \exists y \in T : d(x,y) \leq 2\Phi^{OPT}_{kcenter}(P,k)$

and $\exists c \in S : d(y,c) \leq 2\Phi^{OPT}_{kcenter}(P,k)$

$\Rightarrow d(x,S) \leq d(x,c) \leq d(x,y) + d(y,c)$ (by triangle inequality)

Proof of Theorem

$$\le 4\,\Phi^{OPT}_{kcenter}(P,k)$$



$$d(x,S) \le 4\,\Phi^{OPT}_{kcenter}(P,k)$$

$\square$

Observations on MR-Farthest-First Traversal

- Farthest-First Traversal provides good coresets T_i's, hence a good final coreset T since it ensures that any point not belonging to T is well represented by some coreset point.
- MR-Farthest-First Traversal is able to handle very large pointsets and the the final approximation is not too far from the best achievable one.
- By selecting $k' > k$ centers from each subset P_i in Round 1, the quality of the final clustering improves. In fact, it can be shown that when P satisfy certain properties and k' is sufficiently large, MR-Farthest-First Traversal can almost match the same approximation quality as the sequential Farthest-First Traversal, while, still using sublinear local space and linear aggregate space.

Coreset-based max pairwise distance approximation

Let's recall the problem: given a set S of N points from some metric space determine the maximum distance between any two points of S, namely

$$d_{\max} = \max_{x,y \in S} d(x, s).$$

In a previous lecture, we learned that the maximum distance between an arbitrary point (e.g., x_0) and all other points is at least $d_{\max}/2$, that is, a 2-approximation.

Can we do better?

Coreset-based max pairwise distance approximation

Coreset-based approximation:

- Fix a suitable $k \geq 2$.
- Extract a coreset $S' \subset S$ of size k by running a k-center clustering algorithm on S and taking the k cluster centers as set S'.
- Return $d_{S'} = \max_{x,y \in S'} d(x, y)$ as an approximation of $d_{\max}$.

Can the approximation be computed efficiently?

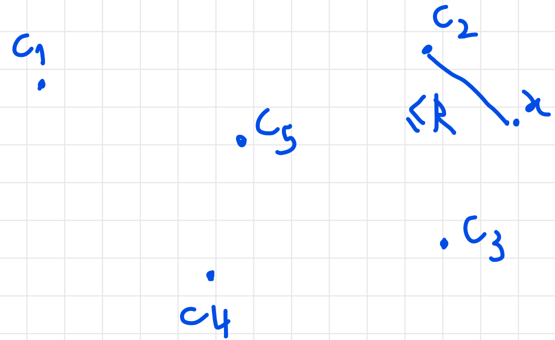
If $k = O(1)$, $d_{S'}$ can be computed

- Sequentially: in $O(N)$ time (using Farthest-First Traversal)
- In MapReduce: in 3 rounds, with local space $M_L = O\left(\sqrt{N}\right)$ and aggregate space $M_L = O(N)$ (using MR-Farthest-First Traversal)

Is $d_{S'}$ a good approximation of $d_{\max}$?

$$S' = \{c_1, c_2, \ldots, c_k\}$$

$k = 5$



$$R \triangleq \max_{x \in S} d(x, S')$$

k-center clustering aims at minimizing R

Let $a, b \in S$ $\quad d(a,b) = d_{max}$

Suppose that a is closest to c_1 and b is closest to c_j

$$d_{max} = d(a,b) \leq d(a,c_1) + d(c_1,c_j) + d(c_j,b)$$

$$\leq 2R + d_s'$$

$\Rightarrow \quad d_{s'} \geq d_{max} - 2R$

If $R \ll d_{max}$ (e.g. $R \leq \varepsilon\, d_{max}$ for small ε)

then $d_{s'} \approx d_{max}$

In practice. R decreases as k grows longer. It can be shown that for low-dimensional spaces one can make $R \leq \varepsilon \cdot d_{max}$ (small ε) with constant k

Exercise

Let P be a set of N points in a metric space (M, d), and let $T \subseteq P$ be a coreset of $|T| > k$ points such that for each $x \in P$ we have $d(x, T) \leq \epsilon \Phi^{\text{opt}}_{\text{kcenter}}(P, k)$, for some $\epsilon \in (0, 1)$. Let S be the set of k centers obtained by running the Farthest-First Traversal algorithm on T, and let $\mathcal{C}_S$ be the clustering returned by Assign(P, S). Prove an upper bound to $\Phi_{\text{kcenter}}(\mathcal{C}_S)$ as a function of ϵ and $\Phi^{\text{opt}}_{\text{kcenter}}(P, k)$.

Exercise

Let P be a set of points in a metric space (M, d), and let $T \subseteq P$. For any $k < |T|, |P|$, show that $\Phi^{\text{opt}}_{\text{kcenter}}(T, k) \leq 2\Phi^{\text{opt}}_{\text{kcenter}}(P, k)$. Is the bound tight?

Exercise

Let P be a set of N points in a metric space (M, d), and let $\mathcal{C} = (C_1, C_2, \ldots, C_k; c_1, c_2, \ldots, c_k)$ be a k-clustering of P. Initially, each point $q \in P$ is represented by a pair $(\mathrm{ID}(q), (q, c(q)))$, where $\mathrm{ID}(q)$ is a distinct key in $[0, N-1]$ and $c(q) \in \{c_1, \ldots, c_k\}$ is the center of the cluster of q.

1. Design a 2-round MapReduce algorithm that for each cluster center c_i determines the most distant point among those belonging to the cluster C_i (ties can be broken arbitrarily).
2. Analyze the local and aggregate space required by your algorithm. Your algorithm must require $o(N)$ local space and $O(N)$ aggregate space.

Exercise

Let P be a set of N *bicolored points* from a metric space, partitioned into k clusters $C_1, C_2, \ldots, C_k$. Each point $x \in P$ is initially represented by the key-value pair $(\mathrm{ID}_x, (x, i_x, \gamma_x))$, where ID_x is a distinct key in $[0, N-1]$, i_x is the index of the cluster which x belongs to, and $\gamma_x \in \{0, 1\}$ is the color of x.

1. Design a 2-round MapReduce algorithm that for each cluster C_i checks whether all points of C_i have the same color. The output of the algorithm must be the k pairs (i, b_i), with $1 \leq i \leq k$, where $b_i = -1$ if C_i contains points of different colors, otherwise b_i is the color common to all points of C_i.
2. Analyze the local and aggregate space required by your algorithm. Your algorithm must require $o(N)$ local space and $O(N)$ aggregate space.

References

LRU14 J. Leskovec, A. Rajaraman and J. Ullman. Mining Massive Datasets. Cambridge University Press, 2014. Sections 3.1.1 and 3.5, and Chapter 7

BHK18 A. Blum, J. Hopcroft, and R. Kannan. Foundations of Data Science. Manuscript, June 2018. Chapter 7